EXPERIMENT NO: - 02

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AIM: - To study different types of sensors, their working principles, applications, and the wireless technologies used for data transmission.

SENSORS

1) Temperature Sensor

Purpose: -
To monitor and regulate temperature in various environments to ensure safety, efficiency, and performance. It prevents overheating in machines, maintains ideal conditions in HVAC systems, and supports automation in industrial and household applications.
Examples: -
Thermocouples, RTDs, Thermistors, Semiconductor Temperature Sensors (IC-based)
Key Technologies used: -
Thermoelectric Effect, Resistance Variation with Temperature, Infrared Sensors, Semiconductor-Based Sensing
Principle of Working: -
Temperature sensors work by converting temperature changes into electrical signals. Thermocouples generate voltage based on the Seebeck effect, while RTDs and thermistors change their resistance with temperature variations. Infrared sensors detect emitted infrared radiation, and semiconductor sensors use temperature-dependent changes in voltage or current to determine temperature.
Applications: -
Industrial, medical, Food Industry, IOT & Smart homes

2) Humidity Sensor

Purpose: -
To measure and monitor moisture levels in the air for climate control, industrial processes, and health-related applications. It helps maintain optimal humidity in homes, greenhouses, and storage facilities.
Examples: -
Capacitive Humidity Sensors, Resistive Humidity Sensors, Thermal Conductivity Sensor
Key Technologies used: -
Capacitive Sensing, Resistive Sensing, Thermal Conductivity

Principle of Working: -
Humidity sensors detect moisture levels by measuring changes in capacitance, resistance, or thermal conductivity. Capacitive sensors measure humidity by detecting variations in a dielectric material's capacitance. Resistive sensors use hygroscopic materials that change resistance with humidity, while thermal conductivity sensors compare heat conductivity differences between dry and humid air.
Applications: -
Weather monitoring, HVAC systems, agriculture, industrial processes, healthcare

3) Gas Sensor

Purpose: -
To detect and measure the concentration of gases in the air for safety, environmental monitoring, and industrial applications. It helps in detecting harmful gases like CO, CO_2, methane, and toxic fumes
Examples: -
Electrochemical Gas Sensors, Infrared Gas Sensors, Semiconductor Gas Sensors, Catalytic Gas Sensors
Key Technologies used: -
Electrochemical Reaction, Infrared Absorption, Semiconductor Oxide Sensing, Catalytic Combustion
Principle of Working: -
Gas sensors detect gases by utilizing different principles. Electrochemical sensors produce an electrical signal based on a chemical reaction with the target gas. Infrared sensors measure gas absorption at specific wavelengths. Semiconductor sensors change resistance when exposed to gases, and catalytic sensors detect combustible gases through heat-induced oxidation.
Applications: -
Air quality monitoring, industrial safety, gas leak detection, automotive emissions, smart homes

4) Motion Sensor

Purpose: -
To detect movement in an area for security, automation, and tracking applications. It is used in surveillance systems, automatic lighting, and smart devices.
Examples: -
Passive Infrared (PIR) Sensors, Ultrasonic Sensors, Microwave Sensors, Tomographic Motion Sensors

Key Technologies used: -
Infrared Sensing, Ultrasonic Waves, Microwave Doppler Effect, Tomographic Detection
Principle of Working: -
Motion sensors detect movement through changes in infrared radiation, sound waves, or electromagnetic signals. PIR sensors detect body heat movement, ultrasonic sensors measure wave reflections, microwave sensors use the Doppler effect to detect motion, and tomographic sensors monitor disturbances in radio waves.
Applications: -
Security systems, smart lighting, automation, robotics, gesture recognition

5) **Light Sensor**

Purpose: -
To detect and measure light intensity for applications in automation, display brightness control, and environmental monitoring.
Examples: -
Photodiodes, Photoresistors (LDR), Phototransistors, Ambient Light Sensors
Key Technologies used: -
Photoelectric Effect, Light-Dependent Resistance, Semiconductor-Based Light Detection
Principle of Working: -
Light sensors work by converting light into electrical signals. Photodiodes and phototransistors generate current when exposed to light. LDRs (photoresistors) change resistance based on light intensity, and ambient light sensors adjust brightness in screens and displays.
Applications: -
Automatic lighting, smartphones, cameras, energy management, optical communication

6) **Proximity Sensor**

Purpose: -
To detect the presence or absence of an object without physical contact, commonly used in automation, touchless interfaces, and security systems.
Examples: -
Capacitive Proximity Sensors, Inductive Proximity Sensors, Infrared Proximity Sensors, Ultrasonic Proximity Sensors
Key Technologies used: -
Electromagnetic Field Sensing, Capacitive Coupling, Infrared Reflection, Ultrasonic Echo Detection
Principle of Working: -
Proximity sensors work by emitting electromagnetic fields, infrared rays, or sound waves and

detecting changes in reflection or signal disruption. Inductive sensors detect metallic objects, capacitive sensors sense any material, infrared sensors detect reflected IR light, and ultrasonic sensors measure sound wave reflections.

Applications: -

Smartphones, automation, security, automotive parking sensors, industrial robotics

7) **Pressure Sensor**

Purpose: -

To measure the force exerted by a fluid (gas or liquid) on a surface, used in industrial, medical, and automotive applications.

Examples: -

Piezoelectric Pressure Sensors, Capacitive Pressure Sensors, Strain Gauge Pressure Sensors, Optical Pressure Sensors

Key Technologies used: -

Piezoelectric Effect, Strain Gauge Resistance, Capacitive Sensing, Optical Interference

Principle of Working: -

Pressure sensors convert mechanical force into an electrical signal. Piezoelectric sensors generate voltage under pressure, strain gauge sensors change resistance with deformation, capacitive sensors measure changes in capacitance, and optical sensors use light interference to detect pressure variations.

Applications: -

Aerospace, automotive, weather monitoring, medical devices, industrial automation

8) **pH Sensor**

Purpose: -

To measure the acidity or alkalinity of a solution, used in chemical industries, water treatment, and agriculture.

Examples: -

Glass Electrode pH Sensors, ISFET pH Sensors, Optical pH Sensors

Key Technologies used: -

Electrochemical Potential Measurement, Ion-Sensitive Field-Effect Transistors (ISFET), Optical Absorption

Principle of Working: -

pH sensors measure hydrogen ion concentration in a solution. Glass electrode sensors generate voltage based on pH changes, ISFET sensors use semiconductor materials to detect ion activity, and optical pH sensors measure color changes in pH-sensitive dyes.

Applications: -

Water quality monitoring, agriculture, chemical processing, food and beverage industry

9) **Flow Sensor**

Purpose: -
To measure the rate of liquid or gas flow in pipelines for industrial automation, medical devices, and environmental monitoring.
Examples: -
Ultrasonic Flow Sensors, Electromagnetic Flow Sensors, Thermal Flow Sensors, Turbine Flow Sensors
Key Technologies used: -
Ultrasonic Doppler Effect, Electromagnetic Induction, Thermal Sensing, Mechanical Turbine Rotation
Principle of Working: -
Flow sensors detect fluid movement using various methods. Ultrasonic sensors measure the Doppler shift in sound waves, electromagnetic sensors detect voltage changes in conductive fluids, thermal sensors measure heat dissipation, and turbine sensors track rotation speed of a mechanical turbine.
Applications: -
Water management, medical respiratory devices, fuel monitoring, industrial fluid control

WIRELESS TECHNOLOGIES USED

1. **Temperature Sensor**

Wireless Technologies:

- **Wi-Fi** – Used in smart home systems for temperature control and remote monitoring.
- **Zigbee** – Common in industrial temperature monitoring systems, especially in IoT applications.
- **Bluetooth Low Energy (BLE)** – Used in wearable devices and portable temperature sensing systems.
- **LoRa** – Suitable for long-range temperature monitoring in agriculture, outdoor environments, and remote areas.
- **NB-IoT** – Applied in smart city infrastructure and utility monitoring, including temperature sensors in street lights or utility meters.

2. **Humidity Sensor**

   Wireless Technologies:

- **Wi-Fi** – Used in smart home and industrial monitoring systems.
- **LoRa (Long Range)** – Ideal for agriculture and remote environmental monitoring.
- **Bluetooth Low Energy (BLE)** – Used in portable health and weather monitoring devices.
- **Zigbee** – Suitable for IoT-based home automation and greenhouse monitoring.

3. **Gas Sensor**

   Wireless Technologies:

- **LoRa** – Used in smart city air quality monitoring systems.
- **Wi-Fi** – Integrated into smart home gas leak detectors.
- **Zigbee** – Used in industrial safety applications.
- **NB-IoT (Narrowband IoT)** – Ideal for real-time air pollution monitoring.

4. **Motion Sensor**

   Wireless Technologies:

   - **Wi-Fi** – Used in smart security cameras and alarm systems.
   - **Zigbee** – Integrated into home automation and smart lighting.
   - **BLE** – Used in wearable motion-tracking devices.
   - **RFID** – Suitable for asset tracking and security applications.

5. **Light Sensor**

   Wireless Technologies:

   - **Zigbee** – Used in smart lighting control systems.
   - **BLE** – Found in ambient light sensors for smartphones and smart glasses.
   - **Wi-Fi** – Integrated into smart energy management systems.
   - **LoRa** – Used for remote light intensity monitoring in agriculture.

6. **Proximity Sensor**

   Wireless Technologies:

   - **NFC (Near Field Communication)** – Used in contactless payment systems and access control.

- **RFID** – Applied in inventory tracking and security applications.
- **BLE** – Used in mobile proximity sensing and beacon technology.
- **Wi-Fi** – Found in smart home automation and touchless interfaces.

7. **Pressure Sensor**

   Wireless Technologies:

   - **Wi-Fi** – Used in industrial pressure monitoring.
   - **LoRa** – Applied in smart agriculture and remote pipeline monitoring.
   - **Zigbee** – Found in HVAC and medical applications.
   - **NB-IoT** – Used for water pressure and smart grid monitoring.

8. **pH Sensor**

   Wireless Technologies:

   - **LoRa** – Used in water quality monitoring in remote areas.
   - **Wi-Fi** – Applied in smart agriculture and industrial automation.
   - **Zigbee** – Used in smart farming applications.
   - **BLE** – Found in portable pH measurement devices.

9. **Flow Sensor**

   Wireless Technologies:

   - **NB-IoT** – Used in smart water and gas metering.
   - **LoRa** – Ideal for industrial fluid monitoring and irrigation systems.
   - **Wi-Fi** – Used in smart home water leak detection systems.
   - **Zigbee** – Applied in energy-efficient smart city projects.

Conclusion: -

Sensors play a crucial role in detecting, measuring, and transmitting data for automation and control. Different types of sensors operate on specific principles to suit various applications. Wireless advancements enhance their efficiency with real-time data transmission. Future innovations will further expand their accuracy, connectivity, and impact.